

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Branch: Civil Engineering

Semester: VII

Subject Code & Name: BTCVC703 Construction Techniques

Max Marks: 60

Date:10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	_____ consist of a large bucket which is attached to a tractor. a. Bulldozer b. Scraper c. Grader d. Escalator	BL1/C01	1
2	The size of bulldozer is indicated by the dimension of its _____. a. Site b. Tyre c. Engine d. Blades	BL1/C01	1
3	In Vacuum method, the top of bore hole is sealed by _____. a. Soil Cement b. Bentonite c. Clay d. All	BL1/C02	1
4	Which of the earth moving machine has shortest cycle time? a. Drag line b. Hoe c. Calmp Shell d. Dipper Shovel	BL1/C02	1
5	Which process comes after batching in manufacture process? a. Transportation b. Placing c. Mixing d. Compacting	BL1/C03	1
6	Ready mix plant & central mix plant differ in _____. a. Transportation b. Setting Time c. Properties d. Water addition	BL1/C03	1
7	_____ is provided at the top of the tremie to receive concrete. a. Bucket b. Hopper c. Tremie d. Pipe	BL1/C04	1
8	Excavation of soil inside the well in the river beds can be done by _____. a. Auger b. Sengind in workers c. Special equipment d. None	BL1/C04	1
9	Which of the following are used for the horizontal movement of material in a facility. a. Cranes b. Lifters c. Buckets d. Lathes	BL1/C04	1
10	Which of the following is a mechanical device used to raise & lower loads? a. Chains b. Hoists c. Bearings d. Guided Wires	BL1/C04	1

11	The places where there is a passage of flood water then the highway has to be built on _____.	BL1/C04	1
	a. Embankment b. Subway c. Overpass d. Underpass		
12	The layer which is constructed above embankment is called _____.	BL1/C04	1
	a. Subgrade b. Fill c. Base d. Sub base		
Q. 2 Solve the following.			
A)	Differentiate Mechanical & Manual construction.	BL4/C01	6
B)	Explain about the equipment's involved in the construction of multistoried building.	BL2/C01	6
Q.3 Solve the following.			
A)	Explain what is ripper? Enlist various factor affecting output of ripper.	BL2/C02	6
B)	Describe dewatering? Explain in detail various methods of dewatering.	BL2/C02	6
Q. 4 Solve Any Two of the following.			
A)	Explain RMC? Explain the working of RMC plant with diagram?	BL2/C03	6
B)	Describe under water concreting? Explain in detail working of any two methods with neat sketch.	BL2/C03	6
C)	Write a short note on Grouting.	BL2/C03	6
Q.5 Solve Any Two of the following.			
A)	Describe What is Prefabricated Construction along with its types.	BL2/C04	6
B)	Explain what is mean by dredging? Explain the various types of dredgers?	BL2/C04	6
C)	Enlist the different types of cranes? Explain any three the types in detail?	BL2/C04	6
Q. 6 Solve Any Two of the following.			
A)	Explain the various factors needed to consider in road construction?	BL2/C04	6
B)	How do you build a new railway line? What is the most important strategic reasons for the construction of a new railway line?	BL2/C04	6
C)	Explain diaphragm wall in detail.	BL2/C04	6
*** End ***			